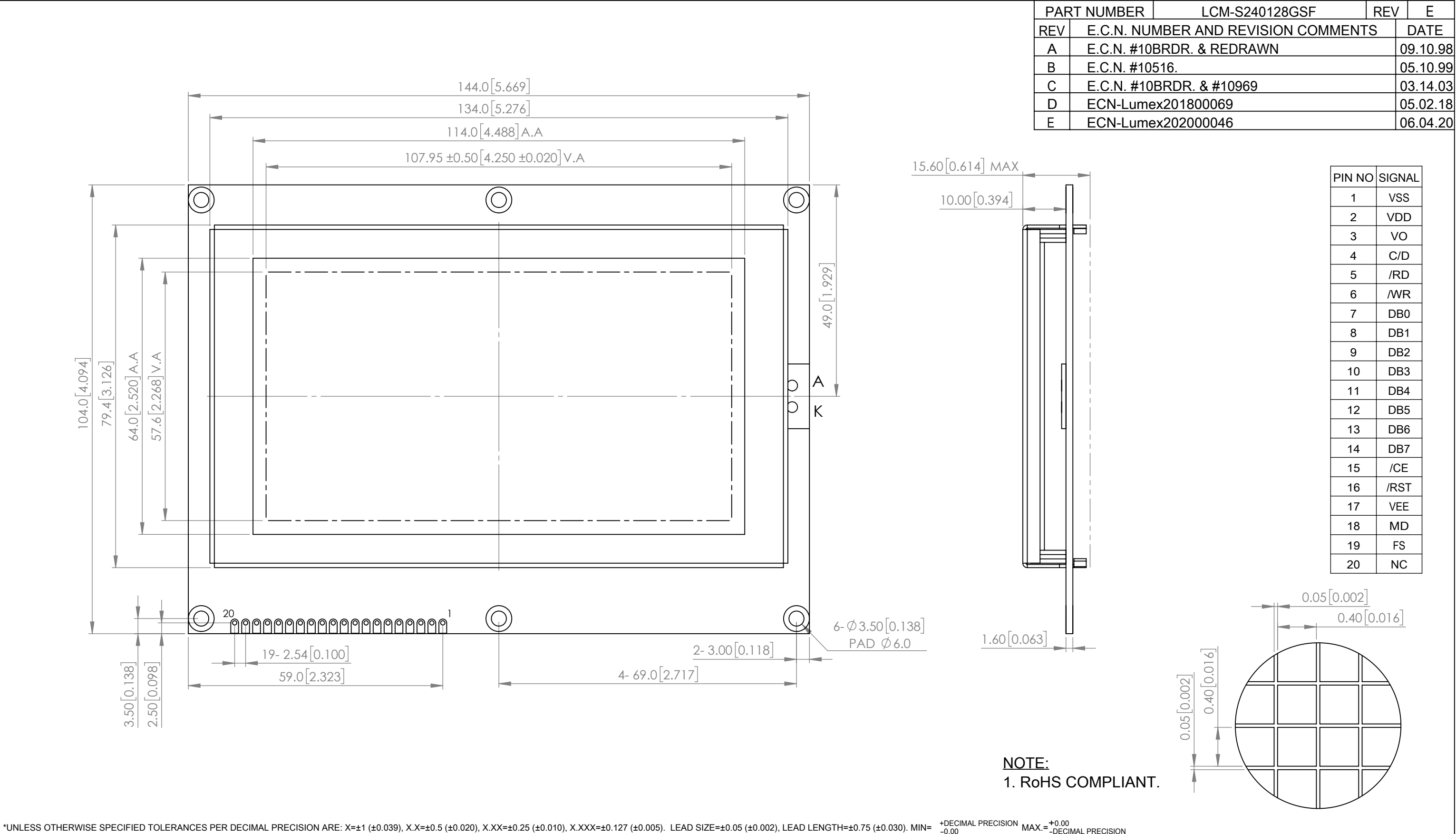




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425 N. GARY AVE.
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144.0(W)×104.0(H)×10.8/15.6(D)MAX. mm, 240x128 DOT MATRIX GRAPHIC LCD MODULE, STN GRAY, POSITIVE, 1/128 DUTY, LED BACKLIGHT.

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PAGE : 1 OF 4

CHKD BY : E.C.

SCALE : NTF

APRVD BY : G.Y.

UNIT : mm [INCH]



			PART NUMBER	LCM-S240128GSF		REV	E
GENERAL SPECIFICATION							
ITEM		DESCRIPTION					
LCD TYPE		STN, POSITIVE					
REAR POLARIZER		TRANSFLECTIVE					
BACKLIGHT TYPE		LED					
BACKLIGHT COLOR		YELLOW GREEN					
TEMPERATURE RANGE		NORMAL TEMP., 5V					
VIEWING DIRECTION		6 O'CLOCK					
VIEWING AREA		114.0 (W)mm x 64.0 (H)mm					
MODULE SIZE		144.0 (W)mm x 104.0 (H)mm x 15.6 MAX(D)mm					
PIXELS SIZE		0.40 x 0.40					
PIXELS PITCH		0.45 x 0.45					
DISPLAY FORMAT		240(W) x 128(H)					
DUTY RATIO		1 / 128 DUTY					
ABSOLUTE MAXIMUM RATING							
(1) Electrical Absolute Ratings							
ITEM		SYMBOL	MIN.	MAX.	UNIT		
POWER SUPPLY FOR LOGIC		VDD-VSS	-0.3	7	VOLT		
POWER SUPPLY FOR LCD		VDD-Vo	-0.3	24	VOLT		
INPUT VOLTAGE		V1	-0.3	VDD	VOLT		
LED POWER DISSIPATION		PAD	-	7590	mW		
LED FORWARD CURRENT		IAF	-	1650	mA		
LED REVERSE VOLTAGE		VR	-	8	V		
(2) Environmental Absolute Maximum Ratings							
ITEM		WIDE TEMPERATURE					
		OPERATING		STORAGE			
		MIN.	MAX.	MIN.	MAX.		
AMBIENT TEMPERATURE		0 °C	+50 °C	-20 °C	+70 °C		
HUMIDITY (WITHOUT CONDENSATION)		NOTE 1, 2		NOTE 3, 4			
Note 1: Ta≤50°C: 80% RH max Ta>50°C: Absolute humidity must be lower than the humidity of 85%RH at 50°C							
Note 2: Background color changes slightly depending on ambient temperature. This phenomenon is reversible.							
Note 3: Ta at -20°C will be<48hrs at 70°Cwill be <120hrs when humidity is higher than 75%.							
Note 4: Ta≤70°C: 75RH max Ta>70°C: absolute humidity must be lower than the humidity of 75%RH at 70°C							
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					SCALE : NTF		APRVD BY : G.Y.
					UNIT : mm [INCH]		

ELECTRICAL CHARACTERISTICS

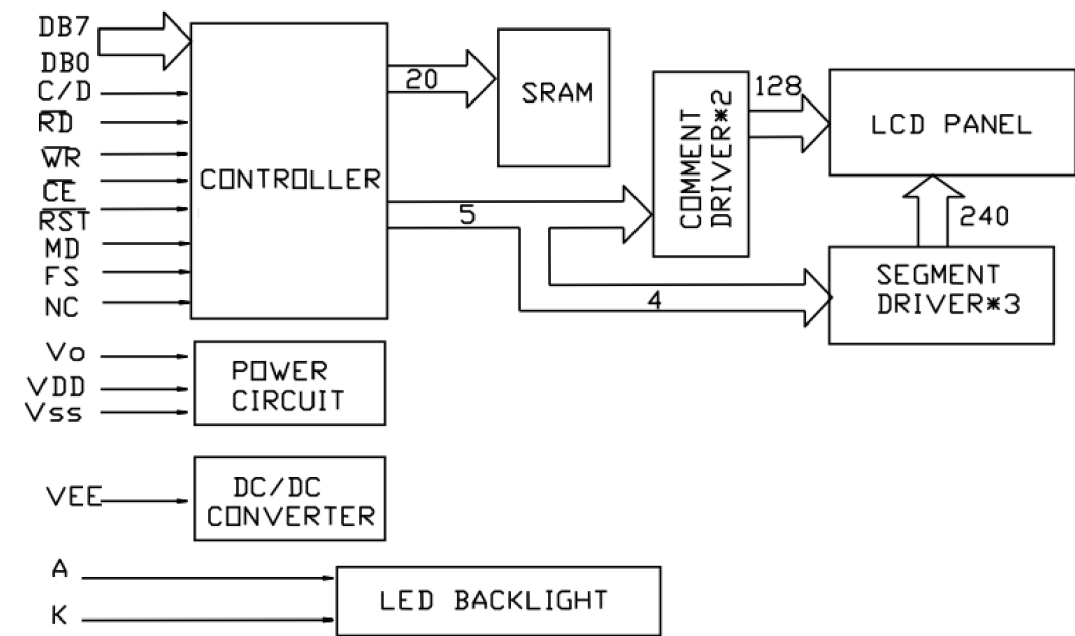
ITEM	SYMBOL	CONDITION	MIN.	TYP.	MAX.	UNIT
POWER SUPPLY FOR LOGIC	VDD-VSS	-	4.5	5.0	5.5	VOLT
POWER SUPPLY FOR LCD	VEE-VSS	-	-14.2	-15.0	-15.9	VOLT
INPUT VOLTAGE	VIL	L LEVEL	0	-	0.8	VOLT
	VIH	H LEVEL	-2.2 VDD	-	VDD	VOLT
	VDD-VO	Ta=25°C	17	18.0	18.6	VOLT
POWER SUPPLY CURRENT FOR LCM	IDD	VDD=5.0V VDD-VO=18.0V	-	17.5	23.0	mA
THE BRIGHTNESS OF BACKLIGHT	L	If =1100mA	-	180	-	cd/m ²
PEAK EMISSION WAVELENGTH	λP	If =1100mA	-	570	-	nm
LED FORWARD VOLTAGE	VF	If=1100mA	-	4.1	4.6	VOLT

ELECTRICAL CHARACTERISTICS

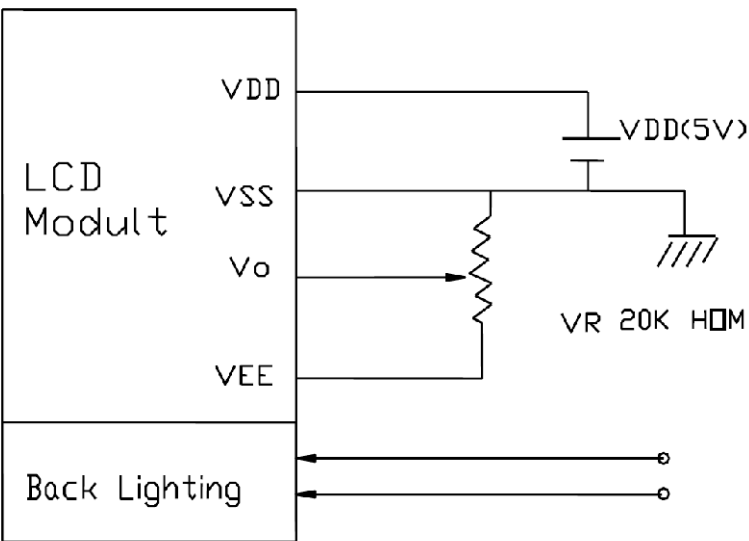
PIN NO.	PIN OUT	FUNCTION DESCRIPTION
1	VSS	Power Supply Ground
2	VDD	Power Supply Voltage
3	Vo	Contrast Adjustment Voltage
4	C/D	WR='L', C/D='H': Command Write WR='L', C/D='L': Data Write RD='L', C/D='H': Status Read RD='L', C/D='L': Data Read
5	/RD	Data Read
6	/WR	Data Write
7	DB0	Data Bit 0
8	DB1	Data Bit 1
9	DB2	Data Bit 2
10	DB3	Data Bit 3
11	DB4	Data Bit 4
12	DB5	Data Bit 5
13	DB6	Data Bit 6
14	DB7	Data Bit 7
15	/CE	Enable Signal
16	/RST	Reset Signal
17	VEE	No Connection
18	MD	H : 32/L : 40 Select of Columns
19	FS	H : 6*8/L : 8*8 Select of Font
20	NC	No Connection

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BLOCK DIAGRAM



POWER SUPPLY



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